Yiyi (Yvonne) Wu

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Education

New York University Sept 2024 – May 2026
Masters in Computer Science GPA: 4.0/4.0
Relevant Courses: Computer Vision; Deep Decision Making and Reinforcement Learning; Learning with Large Language and Vision Models; Reinforcement Learning with Foundation Models

New York University Sept 2019 – May 2023
Bachelor of Arts, Joint Major in Economics and Computer Science GPA: 3.8/4.0

Research

EMERGE Lab, NYU Tandon Feb 2026 - Present
– Advised by Prof. Eugene Vinitsky
– Research focus: Multi-agent World Model, Offline Reinforcement Learning, Multi-agent Self-play

CILVR Lab, NYU Courant Sept 2024 - Present
– Advised by Prof. Lerrel Pinto
– Research focus: Hierarchical World Wodel, Uncertainty Measures

AI & Predictive Analysis Lab, NYU Courant June 2023 - Dec 2024
– Advised by Prof. Anasse Bari
– Research focus: Information retrieval, Retrieval-Augmented Generation

Publications

Zhou, Gaoyue, **Yvonne Wu**, Zichen Jeff Cui, Nicolas Ballas, Mido Assran, Lerrel Pinto, and Yann LeCun (2026). "Hierarchical World Models with Implicit Dynamics". In: *The Fortieth Annual Conference on Neural Information Processing Systems*. [OpenReview].

Ji, Binbin, Siddharth Agrawal, Qiance Tang, and **Yvonne Wu** (2025). "Enhancing spatial reasoning in vision-language models via chain-of-thought prompting and reinforcement learning". In: *arXiv preprint arXiv:2507.13362*. [arXiv].

Bari, Anasse, Tanya Pushkin Garg, **Yvonne Wu**, Sneha Singh, and David Nagel (2024). "Exploring artificial intelligence techniques to research low energy nuclear reactions". In: *Frontiers in Artificial Intelligence* 7, p. 1401782. [Paper].

Bari, Anasse, Sneha Singh, **Yvonne Wu**, Tanya Pushkin Garg, Emos Ker, Charles Wang, Benjamin Kang, Adelina Simpson, Dongjoo Lee, and David Nagel (2024). "Towards Understanding Low Energy Nuclear Reactions: Structuring the Literature and Applying Data Analytics". In: *2024 9th International Conference on Big Data Analytics (ICBDA)*. IEEE, pp. 70–84. [Paper].

Projects

Compact World Model for Scaling Up Reinforcement Learning Spring 2026 – Present
EMERGE Lab

– Develop compact latent dynamics from egocentric observations for scalable reinforcement learning of autonomous driving policies
– Train dynamics models from egocentric video using minimal supervision signals
– Incorporate 3D geometric priors and pose-conditioned latent alignment to enforce consistency across agent viewpoints
– Design low-dimensional, sparse, and factorized latent dynamics to improve simulation throughput while preserving planning-relevant information

Hierarchical World Models with Implicit Dynamics Spring 2025 – Spring 2026
CILVR Lab
– Developed a hierarchical world model that separates subgoal generation from local control for long-horizon robotic planning
– Trained high level diffusion world model and evaluated temporal conditioning and subgoal horizons
– Visualized sampled subgoal distributions to diagnose prediction failures and assess subgoal feasibility
– Constrained sampled states during diffusion rollouts to reduce planner exploitation of infeasible subgoals
– Achieved 38.5% improvement over state-of-the-art policy and diffusion-based planners
– Paper accepted to NeurIPS 2026 Poster and presented at WM@Booth 2026

Enhancing Spatial Reasoning in Vision-Language Models Spring 2025
Course Project, Learning with Large Language and Vision Models
– Improved spatial reasoning in VLMs with chain-of-thought prompting and reinforcement learning
– Designed training strategies to align reasoning traces with structured spatial representations
– Evaluated gains on compositional and multi-step visual reasoning benchmarks
– Improved visual information extraction accuracy on CV-Bench by 43.16%

AI Techniques for Low Energy Nuclear Reaction Research Fall 2023 – Fall 2024
AI & Predictive Analysis Lab
– Developed cluster-based semantic retrieval methods to for LENR scientific literature
– Designed information retrieval and RAG pipelines for grounded domain-specific knowledge extraction
– Assessed the effectiveness and scalability of retrieval frameworks in surfacing latent research directions and enabling hypothesis generation

Experience

Latte (Techstars Startup Accelerators), Co-founder & CTO Apr 2022 – May 2023
– Designed and built a mobile app to support higher-education administrators in preparing students for career mentorship programs
– Led a team of 7 in Agile and test-driven development of an iOS app
– Optimized and streamlined the technology stack to improve scalability and reliability
– Improved efficiency via code organization and automated workflows, reducing codebase size by 20%

Teaching

Recitation Leader, CS102 Data Structures Fall 2024, Spring 2025, Fall 2025
Teaching Assistant, CS-GA Predictive Analytics Fall 2024, Fall 2025
Teaching Assistant, CS-GA Big Data Science Spring 2025

Awards & Honors

2026	Hedwig Kurzbart Master's Fellowship for academic achievement in the Computer Science Department
2024	Computer Science Prize for Academic Excellence and Service to the Department
2023	Cum Laude
2019–2023	Dean's List
2021	Julius Silver Scholarship